WEEKLY PROJECT PROGRESS REPORT

Project Title: A1Cognito – Detection of HbA1c-Glycemic Discordance Using Machine Learning Approaches from Routine Clinical and Haematological Parameters.

Project Guide	**Dr. Pooja Agarwal**
Team Members	**Shivanand** **Shreyash M. Yadwad** **Shivangouda I. Patil** **Samartha M. S.**
Reporting Period	**August and September**

AUGUST — WEEKLY PROGRESS

Week 1 — Problem Identification

- Discussed HbA1c–glycemic discordance and why laboratory HbA1c may not always match a person's glucose exposure.
- Defined the broad project objective: explore a data-driven, explainable approach to screening for potential discordance using routine clinical and haematological parameters.

Output: Initial problem definition and project direction.

Week 2 — Literature Review, Eight Categories and Project Scope

- Reviewed relevant literature and organized factors that can affect HbA1c–glycemic discordance into eight categories.
- Selected five major categories—Categories 2, 4, 5, 6 and 7—as the scope of A1Cognito.
- Used these selected categories to guide the literature review and identify relevant routine clinical, biochemical, demographic and haematological variables.

Output: Eight-category taxonomy identified; five categories selected as the project scope.

Week 3 — Literature Survey and Research Gap

- Compared published work on personalized HbA1c–CGM/GMI modelling, clinical checklists for discordance, and clinical conditions that can make HbA1c unreliable.
- Reviewed research related to ferritin, metabolic variables, anaemia and HbA1c interpretation.
- Summarized limitations of existing approaches and how A1Cognito differs by targeting routine-data-based screening and explainable risk prediction.

Output: Literature survey and preliminary research-gap summary.

Week 4 — NHANES Dataset Acquisition, Processing and Merging

- Downloaded relevant NHANES component files in XPT format, selecting files and variables that corresponded to the project's five chosen categories.
- Converted the required XPT files into CSV format for data inspection and processing.
- Merged the selected data components using participant IDs so that readings from different NHANES files were aligned for the same individual.
- Cleaned the merged data and retained participants with relevant readings for the selected categories, resulting in a dataset of 8,385 patients and 22 features/variables per patient.

Output: Cleaned, merged dataset prepared with 8,385 patients and 22 features.

SEPTEMBER — WEEKLY PROGRESS

Week 1 — System Flowchart Development

- We initiated team meetings to discuss and finalize the overall A1Cognito system flowchart suggested by our guide.
- Organized the main modules: input, medical knowledge, data engineering, AI/ML, explainability/validation/deployment, and output.
- Refined the flow of information between modules, including data preparation, discordance labelling, risk prediction, explanations and validation checkpoints.

Output: System flowchart prepared and module relationships documented.

Week 2 — Research Paper Review and Literature Survey Finalization

- Read and discussed five research papers relevant to HbA1c reliability, HbA1c–CGM/GMI discordance, anemia, ferritin and clinical assessment:
- Cichosz et al., “Narrowing the A1c gap: Personalized modeling of HbA1c–continuous glucose monitor discordance in type 1 diabetes” — PLOS Digital Health (2026).
- Lenters-Westra et al., “Managing discordance between HbA1c and glucose management indicator” — Diabetic Medicine (2025; Wiley).
- Wu et al., “Glycemic assessment when hemoglobin A1c is unreliable: a clinical-laboratory framework for kidney disease, anemia, and hemoglobinopathies” — Frontiers in Endocrinology (2026).
- Hao et al., “The Association Between METS-IR and Serum Ferritin Level in United States Female: A Cross-Sectional Study Based on NHANES” — Frontiers in Medicine (2022).
- Alzahrani et al., “The effect of different types of anemia on HbA1c levels in non-diabetics” — BMC Endocrine Disorders (2023; Springer Nature).
- Summarized the objectives, key findings, limitations and relevance of the papers for A1Cognito and organized the references for the Review 1 presentation.

Output: Five-paper literature survey reviewed and summarized.

Week 3 — Initial Review 1 PPT Draft

- Roughly created the initial Review 1 presentation slides while following the presentation template provided.
- Organized preliminary content for the project problem, background, literature survey, research gaps and other required review sections.
- Established an initial slide structure and presentation flow for team review.

Output: Initial draft of the Review 1 PPT prepared using the college template.

Week 4 — PPT Revisions and Final Review 1 Presentation

- Reviewed and revised the presentation through two to three versions, making changes to slide content, wording, layout and consistency.
- Followed the final presentation workflow: (1) Problem Statement, (2) Background, (3) Literature Survey, (4) Research Gaps, (5) Scope and Feasibility, (6) Datasets, (7) Applications/Use Cases, (8) Capstone Deliverables, (9) Timeline/Gantt Chart, and (10) References.
- Checked that the final slide sequence presented the problem and motivation first, then the supporting research and gaps, followed by scope, data, applications, deliverables, timeline and references.
- Each team member also prepared individually to present the PPT, reviewing their assigned content and practising how to explain the relevant slides during Review 1.
- Consolidated the revisions into the final Review 1 PPT uploaded for the review.

Output: Final Review 1 PPT completed using the agreed workflow; individual presentation preparation completed.

Project Guide Signature: Date: _____

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